The most important question in twenty-first-century economics may well be what to do with all the superfluous people. What will conscious humans do, once we have highly intelligent non-conscious algorithms that can do almost everything better?

Throughout history the job market was divided into three main sectors: agriculture, industry and services. Until about 1800, the vast majority of people worked in agriculture, and only a small minority worked in industry and services. During the Industrial Revolution people in developed countries left the fields and herds. Most began working in industry, but growing numbers also took up jobs in the services sector. In recent decades developed countries underwent another revolution, as industrial jobs vanished, whereas the services sector expanded. In 2010 only 2 per cent of Americans worked in agriculture, 20 per cent worked in industry, 78 per cent worked as teachers, doctors, webpage designers and so forth. When mindless algorithms are able to teach, diagnose and design better than humans, what will we do?

This is not an entirely new question. Ever since the Industrial Revolution erupted, people feared that mechanisation might cause mass unemployment. This never happened, because as old professions became obsolete, new professions evolved, and there was always something humans could do better than machines. Yet this is not a law of nature, and nothing guarantees it will continue to be like that in the future. Humans have two basic types of abilities: physical abilities and cognitive abilities. As long as machines competed with us merely in physical abilities, you could always find cognitive tasks that humans do better. So machines took over purely manual jobs, while humans focused on jobs requiring at least some cognitive skills. Yet what will happen once algorithms outperform us in remembering, analysing and recognising patterns?

The idea that humans will always have a unique ability beyond the reach of non-conscious algorithms is just wishful thinking. The current scientific answer to this pipe dream can be summarised in three simple principles:

1. Organisms are algorithms. Every animal – including *Homo sapiens* – is an assemblage of organic algorithms shaped by